

Tanay Pratap Singh

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EDUCATION

Master's of Science in Applied Data Science	Syracuse University, Syracuse, NY	Jan. 2025 – Dec. 2026
B.Tech in Computer Science and Engineering	JSSATE Noida, UP, India	Dec. 2020 – May 2024

EXPERIENCE

Research Assistant <i>Syracuse University</i>	Feb. 2026 – Present <i>Syracuse, NY</i>
- Developed data ingestion workflow with Python, Pandas, and Zooniverse API to parse 276,000+ classifications, reconciling 5 schemas across 25,104 glitch subjects and 8,293 channels. - Engineered image processing workflow using OpenCV to retrieve 49,000 spectrograms via batch API calls, transforming 1200x1200 mosaic images into paired 224x224 channel heatmaps for deep learning input. - Trained two-input Convolutional Neural Network with shared MobileNetV2 backbone in TensorFlow/Keras to predict volunteer similarity labels from paired spectrograms; achieved validation AUC of 0.89 on 2,000-subject holdout set. - Applied model calibration techniques (label smoothing, AdamW optimizer, AUC-based early stopping) to reduce Expected Calibration Error from 0.10 to 0.05; generated glitch-type by auxiliary-channel similarity matrix to identify detector sub-systems linked to specific glitch morphologies.	
Data and Product Analyst Intern <i>ProProfs</i>	Jul. 2023 – Aug. 2023 <i>Noida, UP, India</i>
- Performed product and customer analytics using Python and Excel, delivering SWOT, competitive benchmarking (100+ features vs. Zoho and Salesforce), and ICP analyses across SaaS products serving 5,000+ users. - Analyzed customer datasets of 2,000 clients using Python and Excel pivot tables to identify behavior patterns and conversion drivers, enabling the product team to prioritize high-impact features that increased trial-to-paid conversion. - Created interactive Tableau dashboards from SQL and Excel data to visualize feature usage, customer segmentation, and conversion funnels, helping stakeholders identify under-used features and reduce churn by 5 percent.	

PROJECTS

ShopFloor AI	Jul. 2026
* Architected predictive maintenance system using Python, scikit-learn, and MLflow on 10,000 manufacturing sensor records with 5 failure modes, engineering 16 features including power output, thermal differential, and overstrain metrics; best model (Gradient Boosting) achieved 0.85 F1 score and 0.96 AUC-ROC. * Implemented real-time streaming architecture using Apache Kafka for sensor data, computed rolling KPIs with programmatic anomaly detection, and integrated RAG using LangChain, ChromaDB, and OpenAI API to enable natural-language search over manufacturing documentation. * Deployed 7-service containerized application with Docker Compose: FastAPI (REST API), Streamlit (interactive dashboard), Apache Airflow (automated daily retraining with data quality gates), and MLflow (model registry promotion).	
JMA Wireless Production Line Optimization	Jan. 2026 – May 2026
* Directed DMAIC engagement for JMA Wireless, consolidating production data from SAP, ADP, PLC automation, and inspection databases to standardize KPI reporting across two parallel assembly lines, projecting 355,000 dollars in annualized cost recovery. * Built Python ETL pipeline using Pandas and openpyxl to flatten 31 GB of nested PLC JSON data into a 5 MB Power BI ready dataset, eliminating a 3 to 5 day manual reconciliation lag and saving approximately 48 analyst hours per week. * Delivered interactive Power BI dashboard with DAX measures tracking four standardized KPIs; conducted SIPOC, FMEA, and Pugh Matrix analyses to identify root causes and prioritize improvements.	

TECHNICAL SKILLS

Languages and Frameworks: Python, SQL, R, JavaScript, HTML/CSS, FastAPI, REST API, Streamlit
Machine Learning and AI: Scikit-learn, TensorFlow, Keras, XGBoost, CatBoost, LangChain, OpenCV, SHAP, LLM, Generative AI, RAG, Prompt Engineering; Random Forest, Gradient Boosting, SVM, LSTM, CNN, NLP, Deep Learning, Ensemble Methods, Classification, Predictive Modeling, Feature Engineering, Cross-Validation, Model Deployment
Data and Analytics: Pandas, NumPy, Spark, Tableau, Power BI, DAX, Power Query, Excel, Plotly, Matplotlib; EDA, Data Visualization, Data Wrangling, A/B Testing, Time Series Forecasting, Hypothesis Testing, Anomaly Detection, Cohort Analysis, Funnel Analysis, Customer Segmentation
MLOps, Cloud and Data Engineering: Docker, Apache Kafka, Apache Airflow, MLflow, Model Registry, ChromaDB, Vector Databases, OpenAI API, Azure, GCP, MySQL, BigQuery, ETL Pipelines, Data Pipelines, Git/GitHub, Jupyter
Statistics, Modeling and Business: Linear Regression, Logistic Regression, ANOVA, PCA, Lean Six Sigma, DMAIC, SIPOC, Pugh Matrix, Process Mapping, Process Improvement, KPI Design, Business Intelligence, Stakeholder Communication